

Order of operations

- The order operations (specifically, when the mode conversion, propagation, and geometric spreading factors are applied) has an impact on the exact amplitude and phase calculated
- LWPC calculates amplitude and phase for each distance from the beginning of the waveguide agnostic of what has been calculated for any other point
- Significant improvements have been made to matching the calculated fields by executing the same order of operations as LW_SUM_MODES

Step by step

1. Run LWPC with specified settings to calculate necessary
2. From LWPC log file, read in eigenangles, height gain functions, slab parameters, T1-T4, and mode conversion coefficients $A(m,k)$
3. Calculate field values for each desired distance by propagating modes forward and summing at the desired distance

Following the conceptual steps of TD 400 (Ferguson and Snyder, 1980), and the implementation of Fortran subroutines LW_VS_D and LW_SUM_MODES

LWPC Output flags

- "print-mds "

- 0: minimum print from MDS file
- 1: adds attenuation rate, etc.
- 2: adds conversion coefficients
- 3: adds slab integrals

- "print-swg "

- 0: none
- 1: path parameters and mode lists
- 2: mode parameter outputs
- 3: extrapolation parameter outputs

- "print-wf "

- 0: minimum print
- 1: adds iterations
- 2: adds fields vs. height

- "print-lwf "

- 0: minimum print from LWF file
- 1: adds fields vs. distance
- 2: adds mode parameters vs. distance

- "print-mc"

- 0: minimum print
- 1: adds conversion coefficients
- 2: adds integrals

Slab parameters

- Current slab parameters: distance from transmitter (rho, lat, lon), magnetic field (azim, dip, bfield), ground (sigma, epsr), ionosphere(beta, hpr)
- Eigenangles and parameters used to calculate excitation factors given for each mode at slab boundary

T1, T2, T3, T4 and f(R)

mds	lat	lon	rho	azim	dip	bfield	sigma	epsr	beta	hpr	ndx				
16	44.5	67.7	40.	76.9	71.3	0.534	4.E+00	81.	0.30	74.0	0.				
m	EIGENr	EIGENi													
1	89.5214	-4.0702	4.19E-03	-1.46E-02	-1.21E-13	-2.69E-14	3.05E-08	3.62E-08	7.99E-01	2.70E-01	-1.73E-06	2.57E-06			
2	88.4451	-3.1908	7.95E-05	-1.17E-04	-3.63E-11	-2.21E-11	-3.09E-08	-7.85E-08	8.01E-01	2.68E-01	3.36E-04	-4.93E-04			
3	83.9024	-0.9832	1.87E-03	-3.42E-02	-2.14E-13	-6.46E-13	4.70E-08	1.60E-07	8.02E-01	2.55E-01	-4.58E-06	1.62E-06			
4	81.1256	-1.5300	2.45E-04	-1.14E-04	-2.31E-10	-9.92E-11	-4.55E-08	-2.80E-07	8.08E-01	2.44E-01	2.84E-04	-1.01E-03			
5	77.6578	-1.1924	1.30E-03	-2.99E-02	-3.46E-12	-8.21E-12	1.72E-07	5.35E-07	8.15E-01	2.22E-01	-1.76E-05	6.51E-06			
6	75.5969	-2.0055	8.05E-04	-3.21E-04	-5.01E-10	-1.89E-10	-9.84E-08	-7.30E-07	8.29E-01	2.11E-01	2.07E-04	-8.24E-04			
7	71.9142	-1.4472	-4.29E-04	-2.71E-02	-4.42E-11	-6.63E-11	6.23E-07	1.45E-06	8.47E-01	1.78E-01	-5.39E-05	2.22E-05			
8	70.7101	-2.3856	2.60E-03	-1.54E-03	-8.02E-10	-1.52E-10	-4.52E-07	-1.61E-06	8.64E-01	1.76E-01	1.42E-04	-5.35E-04			
9	66.3109	-1.5948	-3.97E-03	-1.90E-02	-4.38E-10	-2.84E-10	2.08E-06	2.60E-06	9.02E-01	1.32E-01	-1.53E-04	7.73E-05			
10	65.7605	-2.5265	6.16E-03	-8.78E-03	-9.70E-10	7.64E-11	-1.88E-06	-2.78E-06	9.16E-01	1.39E-01	1.12E-04	-2.92E-04			
11	60.7845	-1.6095	-2.95E-03	-9.80E-03	-1.33E-09	-3.63E-10	2.83E-06	2.51E-06	9.77E-01	9.29E-02	-3.15E-04	1.94E-04			
12	60.3113	-2.8338	5.54E-03	-1.78E-02	-7.96E-10	1.14E-10	-2.67E-06	-2.81E-06	9.90E-01	1.08E-01	1.01E-04	-1.82E-04			
13	55.1580	-1.6375	-2.18E-03	-6.61E-03	-2.18E-09	-4.32E-10	2.94E-06	2.44E-06	1.06E+00	6.87E-02	-4.65E-04	2.92E-04			
14	54.4715	-3.3554	5.10E-03	-2.13E-02	-7.86E-10	9.78E-11	-2.80E-06	-2.88E-06	1.07E+00	8.92E-02	9.81E-05	-1.55E-04			
15	49.2439	-1.7219	-2.17E-03	-5.86E-03	-3.04E-09	-5.74E-10	3.22E-06	2.59E-06	1.13E+00	5.69E-02	-5.67E-04	3.40E-04			
16	42.8457	-1.8742	-2.50E-03	-6.34E-03	-3.96E-09	-7.85E-10	3.77E-06	3.01E-06	1.18E+00	5.21E-02	-6.13E-04	3.52E-04			

Height Gain Functions

- Relative complex field strengths as a function of height are given for each mode in each slab

Slab number

WF_DRIVER: Output for segment 1

Mode eigenangle

WF_DRIVER: azimuth=308.638 eigen= 89.4216 -4.2619 at h= 0.

WF_DRIVER: eigen= 84.1479 -0.4190 at h=50.

WF_DRIVER: nriter= 1 eigen= 84.1482 -0.4191 f= 4.89E-06

WF_DRIVER: wfht1, wfht2, lev1= 90.000000 43.125000 26

WF_DRIVER: azimuth=308.638 eigen= 84.1482 -0.4191 at h=50.

WF_DRIVER: Field strengths

Fields are normalized to $H_y(z=0)$

Height in km

ht	mag	-- ex	-- ang	mag	-- ey	-- ang	mag	-- ez	-- ang	mag	-- hx	-- ang	mag	-- hy	-- ang	mag	-- hz	-- ang
0.0	1.184E-02	-2.357		9.735E-05	2.974		1.011E+00	3.141		8.222E-03	2.190		1.000E+00	-0.000		9.684E-05	2.973	
1.9	7.653E-03	-2.698		8.218E-03	-2.531		1.004E+00	-3.135		8.242E-03	2.189		9.944E-01	0.007		8.175E-03	-2.532	
3.8	5.609E-03	2.926		1.640E-02	-2.528		1.003E+00	-3.129		8.296E-03	2.187		9.935E-01	0.014		1.632E-02	-2.529	
5.6	6.521E-03	2.242		2.466E-02	-2.528		1.006E+00	-3.124		8.375E-03	2.183		9.967E-01	0.018		2.453E-02	-2.529	
7.5	8.843E-03	1.873		3.300E-02	-2.529		1.012E+00	-3.121		8.468E-03	2.178		1.003E+00	0.021		3.282E-02	-2.530	
9.4	1.127E-02	1.677		4.143E-02	-2.531		1.021E+00	-3.119		8.568E-03	2.172		1.013E+00	0.023		4.122E-02	-2.531	
11.2	1.341E-02	1.551		4.996E-02	-2.533		1.032E+00	-3.119		8.664E-03	2.164		1.025E+00	0.023		4.970E-02	-2.534	
13.1	1.513E-02	1.454		5.858E-02	-2.536		1.046E+00	-3.120		8.748E-03	2.155		1.039E+00	0.022		5.828E-02	-2.537	
15.0	1.638E-02	1.368		6.728E-02	-2.539		1.061E+00	-3.123		8.808E-03	2.145		1.054E+00	0.019		6.693E-02	-2.540	
16.9	1.715E-02	1.283		7.601E-02	-2.543		1.076E+00	-3.127		8.836E-03	2.133		1.070E+00	0.015		7.562E-02	-2.544	
18.8	1.745E-02	1.191		8.475E-02	-2.548		1.091E+00	-3.132		8.822E-03	2.119		1.086E+00	0.010		8.431E-02	-2.548	
20.6	1.733E-02	1.084		9.345E-02	-2.553		1.106E+00	-3.139		8.755E-03	2.104		1.102E+00	0.003		9.296E-02	-2.553	
22.5	1.685E-02	0.954		1.020E-01	-2.558		1.120E+00	3.136		8.627E-03	2.087		1.116E+00	-0.005		1.015E-01	-2.559	
24.4	1.618E-02	0.790		1.105E-01	-2.564		1.132E+00	3.127		8.428E-03	2.067		1.129E+00	-0.014		1.099E-01	-2.565	

Mode Conversion Coefficients

- Complex-valued conversion coefficients are displayed as a matrix $A(m,k)$, where m is the index of the mode in slab 2 and k is the index of the mode in slab 1

Slab number

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LW_STEP: Conversion coefficients for segment 2
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A(1, 1)=	9.69390E-01	3.48940E-02	20*log10(T)=	-0.264	2.062
A(2, 1)=	6.08233E-03	-6.88783E-04	20*log10(T)=	-44.263	-6.461
A(3, 1)=	1.35130E-02	-1.96092E-02	20*log10(T)=	-32.463	-55.429
A(4, 1)=	-1.26151E-03	9.40697E-05	20*log10(T)=	-57.958	175.735
A(5, 1)=	3.62578E-03	-6.72838E-03	20*log10(T)=	-42.335	-61.681
A(6, 1)=	2.40857E-03	2.09844E-03	20*log10(T)=	-49.912	41.064
A(7, 1)=	-1.46057E-03	-1.07507E-03	20*log10(T)=	-54.829	-143.645
A(8, 1)=	1.77852E-03	-3.00692E-03	20*log10(T)=	-49.135	-59.397
A(9, 1)=	1.26132E-03	-6.63547E-05	20*log10(T)=	-57.971	-3.011
A(10, 1)=	3.92172E-04	4.26081E-05	20*log10(T)=	-68.080	6.201
A(11, 1)=	-4.36032E-04	-7.53297E-04	20*log10(T)=	-61.206	-120.064
A(12, 1)=	9.92648E-04	-1.35510E-03	20*log10(T)=	-55.495	-53.776
A(13, 1)=	2.65750E-04	-4.56254E-04	20*log10(T)=	-65.547	-59.781
A(14, 1)=	4.83859E-04	6.65702E-04	20*log10(T)=	-61.692	53.989
A(15, 1)=	1.79408E-05	-6.85862E-04	20*log10(T)=	-63.272	-88.502
A(16, 1)=	6.11161E-04	-2.00586E-04	20*log10(T)=	-63.833	-18.170
A(17, 1)=	6.58754E-04	2.65827E-04	20*log10(T)=	-62.970	21.976

Real

Imag

Calculating Mode Sum

- To calculate the total field for a given field component (e.g. E_z) at a distance ρ ...
- In the first slab, E_z due to a single mode is given by

$$E_z(\theta_n, \rho) = - \underbrace{T 1(\theta_n) \sin(\theta_n)}_{\text{Excitation}} hgr(z, \theta_n) \underbrace{\frac{682.2408 \sqrt{f * power}}{\sqrt{\sin(\rho/r_{earth})}}}_{\text{Geometric Spreading}} \underbrace{e^{-ik\rho(\sin(\theta_n)-1)}}_{\text{Propagation}}$$

- Where θ_n is the eigenangle of the mode, $hgr(z)$ is the height gain function of the mode at height z , f is the transmitter frequency, $power$ is transmitter power, and r_{earth} is the earth's radius
- The total field E_z is then given by the sum of modes

$$E_z(\rho) = \sum_n E_z(\theta_n, \rho)$$

Calculating Mode Sum

- If ρ does not lie in the first slab...
- E_z must first be propagated up to and across each slab boundary as follows:

$$E_{z,0}(\theta_m) = \sum_k -T_1(\theta_k) \sin(\theta_k) hgr(z, \theta_k) e^{-ik(x_2 - x_1)(\sin(\theta_k) - 1)} A(m, k)$$

$x_2 - x_1 = \text{length of previous slab}$

Contribution of the m^{th} mode in the previous slab to the k^{th} mode in the current slab

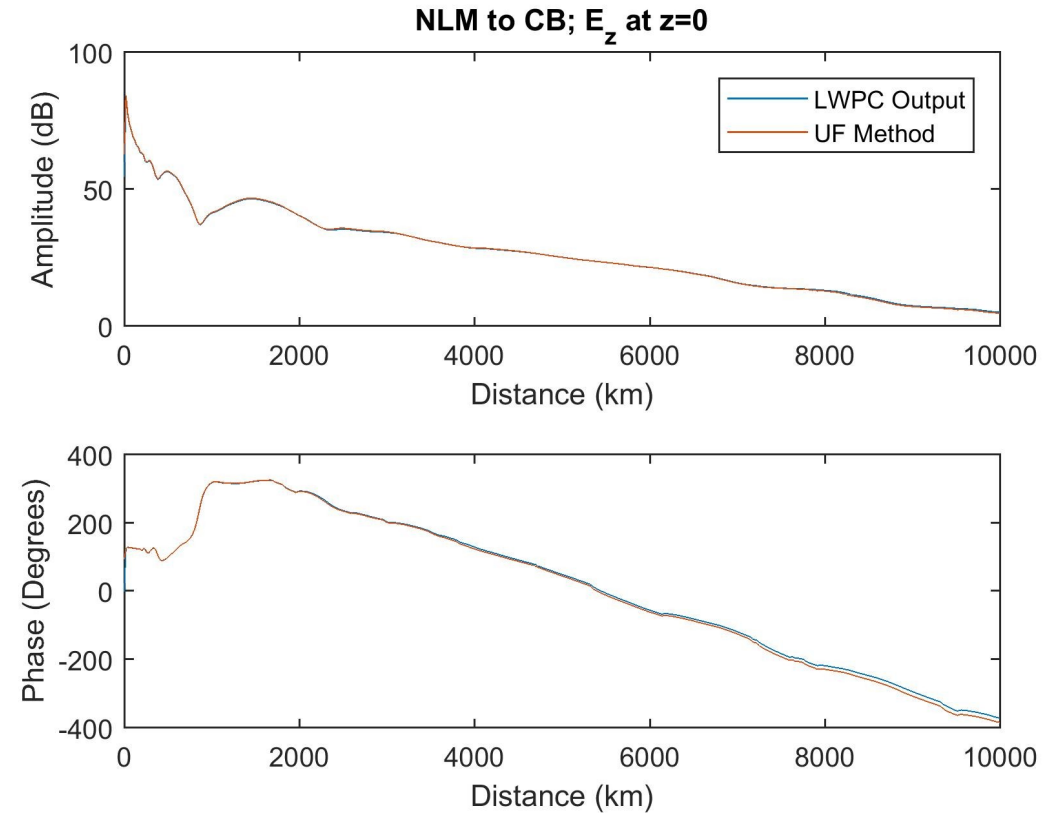
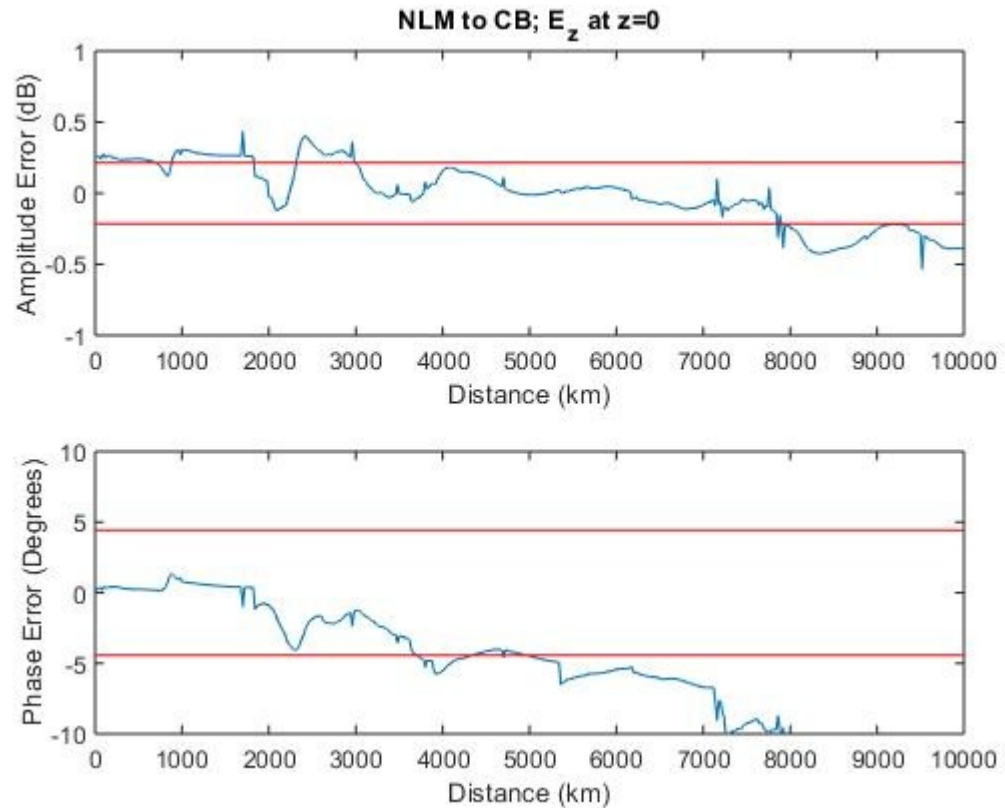
- Where x_2 is the beginning of the current slab and x_1 is the beginning of the previous slab
- The total field E_z is then given by the sum of modes in the final slab with geometric spreading factor applied

Field at beginning of current slab Propagation since beginning of final slab

$$E_z(\rho) = \sum_m E_{z,0}(\theta_m) \frac{682.2408 \sqrt{f * \text{power}}}{\sqrt{\sin(\rho/r_{\text{earth}})}} e^{-ik(\rho - x_2)(\sin(\theta_m) - 1)}$$

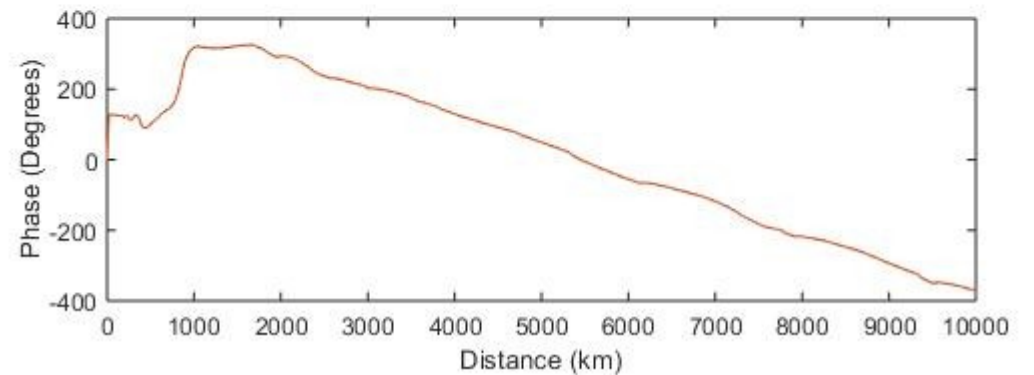
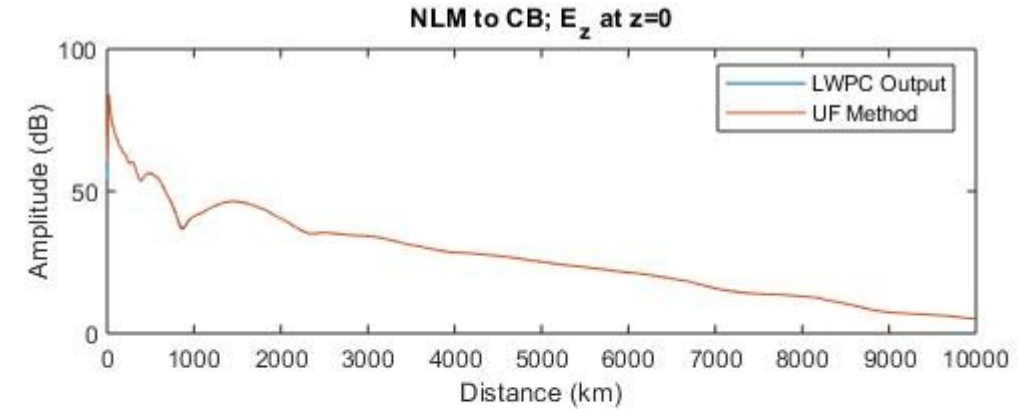
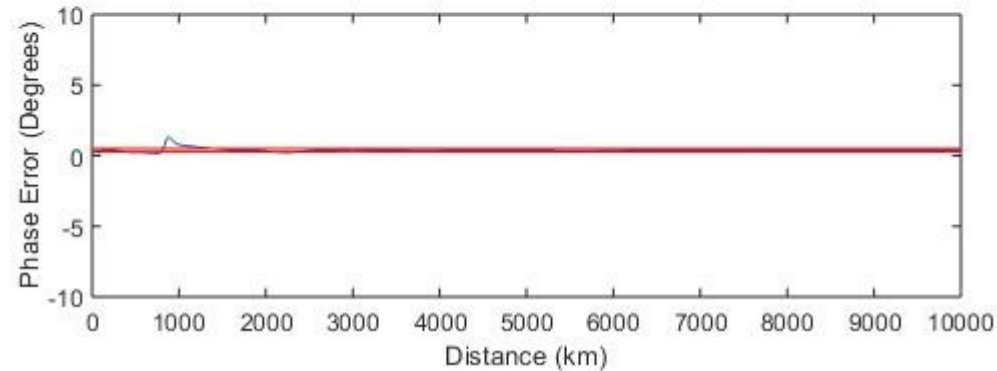
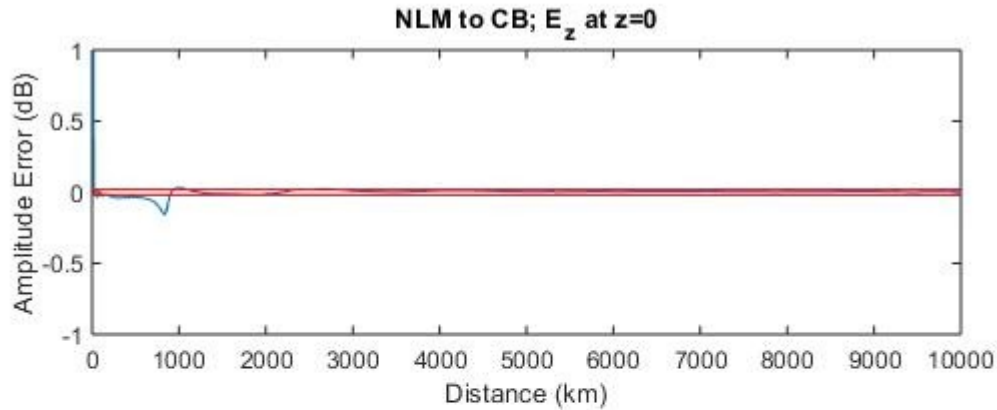
- Any of the 6 field components are calculated this way simply by replacing hgr with the height gain function corresponding to that component and evaluating at all desired heights (z)

Previously Presented Comparison



Error accumulates at large distances

Current Agreement with LWPC



By matching LWPC's order of operations, any accumulated error at distances matches that of LWPC